Module III: Typesetting Mathematics

LaTeX Made Simple · Syllabus units 8–11 · Primer chapter 8 (pages 95–136)

8. Typesetting mathematics

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In plain words. TeX was written to typeset mathematics well. Inside a formula, LaTeX spaces the symbols itself, so you write the meaning and let it place the spaces. Maths has two modes: *in-text* (inside a sentence, with `$...$`) and *display* (a formula on its own line, with `\[...\]` or `equation`).

8.1 The basics

IN-TEXT AND DISPLAYED MATHS

```
The equation representing a straight line in the Cartesian plane is
of the form $ax+by+c=0$, where $a$, $b$, $c$ are constants.

$$
  ax+by+c=0
$$
```

- `$...$` keeps maths in the sentence. The letters are in math italic.
- `$$...$$` (TeX) and `\[...\]` (LaTeX) put the formula on its own centred line.
- `\(...\)` and `\begin{math}...\end{math}` are the LaTeX forms of in-text maths.

Spaces in maths. TeX ignores the spaces you type inside maths and places its own. `$ax+by+c=0$` and `$ax + by + c = 0$` print the same. Do not type spaces to make the result look right.

SUPERSCRIPTS AND SUBSCRIPTS

```
$x^2$      $x_n$      $x_n^2$      % x squared, x sub n, both together
$x^{mn}$    $x_{n-1}$                    % braces group several symbols
$(x^m)^n = x^{mn}$
```

Without braces, only the next character is affected. So `x^mn` prints "x^mn", but `x^{mn}` prints x^{mn}. For a superscript and a subscript together on the same letter, the order and braces change the output: `x_m^n`, `x^n_m` and `{x_m}^n` come out differently. Use braces to say what you mean.

```
$$ x_m^n \qquad x^n_m \qquad {x_m}^n \qquad {x^n}_m $$
```

ROOTS AND FRACTIONS

```
$\sqrt{2}$          % square root
$\sqrt[4]{5}$      % fourth root
$\sqrt{x+y}$       % the top bar stretches over the whole expression
$\sqrt{2+\sqrt{2}}$ % nested roots
```

The square root's horizontal bar stretches to cover its contents. Use `\frac{numerator}{denominator}` for fractions (covered in Part 8.4 below).

SPACING INSIDE MATHS

COMMAND	SPACE	EXAMPLE
`\,`	thin space	`\int f(x)\,dx`
`\;`	thick space	`x_n = x_{n-1}\;(n>2)`
`\quad`	a quad (about the width of M)	`x_1=1,\quad x_2=1`

BINARY OPERATORS AND RELATIONS

TeX adds extra space around *binary operators* (such as `+`, `-`, `\circ`) and around *relations* (such as `=`, `\le`). Symbols that are not in either class do not get that space. To force a symbol to behave as one of these:

```
x\circ y = x+y-xy          % \circ is a binary operator: spaces on both sides
x\mathbin\Box y           % force \Box (from amssymb) to act as a binary operat
x\mathrel\rho y           % force \rho to act as a relation
```

Some symbols (for example `\Box`) are only available after loading `amssymb` or `latexsym`.

8.2 Custom commands

In plain words. If you type the same expression many times, give it a short name with `\newcommand`. Put the definitions in the preamble, so they can be used anywhere.

```
\newcommand{\vect}{(x_1,x_2,\dots,x_n)}
We often write $x$ to denote the vector $\vect$.
```

With an argument, the same command can produce many vectors:

```
\newcommand{\vect}[1]{(#1_1,#1_2,\dots,#1_n)}
$\vect{x}$ gives $(x_1,x_2,\dots,x_n)$ and $\vect{a}$ gives $(a_1,a_2,\dots,a_n

\newcommand{\vect}[2]{(#1_1,#1_2,\dots,#1_#2)}
$\vect{x}{m}$ gives $(x_1,x_2,\dots,x_m)$.
```

`[1]` says the command takes one argument. `#1` is replaced by that argument. A command can take up to nine arguments.

8.3 More on mathematics: the amsmath package

The primer uses `amsmath` for this section. Load it with `\usepackage{amsmath}`. It gives better equation environments.

SINGLE EQUATIONS

ENVIRONMENT	NUMBERED?	USE
`equation*`	No	One display, no number
`equation`	Yes	One display with a number, such as (8.1)
`multline*`, `multline`	No / yes	One long formula over several lines (first line left, last line right)
`split`	Inside another environment	Several lines aligned at `&`, inside one equation

The starred version (`equation*`) has no number. The plain version numbers the formula. For any environment, the unstarred form adds numbers.

```
\begin{equation*}
  (a+b)^2 = (a+b)(a+b)
\end{equation*}
```

Text inside a display uses `\text`. Maths inside that text needs dollar signs:

```
\begin{equation*}
  x\le|x|\quad\text{and}\quad x\ge|x|
\end{equation*}
\begin{equation*}
  x\le|x|\quad\text{for all $x$ in $R$}.
\end{equation*}
```

LONG EQUATIONS: SPLIT AND MULTLINE

When an equation is too long for one line, `split` inside an `equation*` lines it up at the `&` character, with the `=` signs aligned:

```
\begin{equation*}
  \begin{split}
    (a+b)^2 &amp; = (a+b)(a+b)\\
            &amp; = a^2+ab+ba+b^2\\
            &amp; = a^2+2ab+b^2
  \end{split}
\end{equation*}
```

`multline` is simpler to type, but for more than two lines its output is less neat. The primer recommends `split` for long expansions.

GROUPS OF EQUATIONS: GATHER AND ALIGN

```
\begin{gather*}
  (a,b)+(c,d)=(a+c,b+d)\\
  (a,b)(c,d)=(ac-bd,ad+bc)
\end{gather*}

Thus $x$, $y$ and $z$ satisfy the equations
\begin{align*}
  x+y-z &amp; = 1\\
  x-y+z &amp; = 1
\end{align*}
```

- `gather`: each line centred on its own.
- `align`: lines aligned at each `&`. Put the `&` before the `=` to line up the equals signs.
- `\intertext{and by hypothesis}` puts a short piece of text between aligned lines without breaking the alignment.
- Extra `&` signs make side-by-side columns. Use `\qquad` to space the columns apart.
- `aligned`, `gathered` and `alignedat` take only the width they need, so they can sit inside an `equation*` beside other text.

CASES

```
\begin{equation*}
  |x| =
  \begin{cases}
    x      &amp; \text{if $x\ge 0$}\\
    -x     &amp; \text{if $x\le 0$}
  \end{cases}
\end{equation*}
```

EQUATION NUMBERS AND LABELS

```
\numberwithin{equation}{section}   % equation numbers now carry the section num
\begin{equation}\tag{L}
  ax+by+c=0
\end{equation}
\begin{align}
  x+y-z &amp; = 1 \notag \\
  x-y+z &amp; = 1 \tag{H}
\end{align}
```

- Numbers are added to the unstarred environments automatically. In a book or report, the chapter number comes first: the second numbered equation of Chapter 8 prints as (8.2), as in the primer.
- `\numberwithin{equation}{section}` adds the section number as well, in the style (section.equation).
- `\tag{name}` replaces the automatic number with your own label.
- `\notag` removes the number from one row of an `align`.
- Put `\tag` before the `\\` at the end of the row, not after it.

Remember (Part 8.1–8.3)

- Use `$...$` for in-text maths and `\[...\]` or `equation` for displays.
- Braces group the superscript or subscript. `x^{mn}`, not `x^mn`.
- TeX sets the spaces in maths. Do not add spaces by hand.
- Use `align` for several lines that must line up, and `split` inside an equation for one long formula.
- The starred environment is unnumbered; the plain one is numbered.

8 (continued). Matrices, delimiters, fractions, operators and symbols

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8.4 Mathematics miscellany

MATRICES

Matrices are set with environments, where `&` separates the columns and `\\` ends each row:

ENVIRONMENT	BRACKETS
`matrix`	None
`pmatrix`	Round ( )
`bmatrix`	Square [ ]
`vmatrix`	Single bars | | (determinant)
`Vmatrix`	Double bars ‖ ‖
`Bmatrix`	Curly braces { }
`smallmatrix`	None, for use inside a line of text

```
\begin{equation*}
  \begin{pmatrix}
    1 &amp; 1 &amp; -1\\
    1 &amp; -1 &amp; 1\\
    1 &amp; 1 &amp; 1
  \end{pmatrix}
  \begin{pmatrix} x\\ y\\ z \end{pmatrix} =
  \begin{pmatrix} 1\\ 1\\ 1 \end{pmatrix}.
\end{equation*}
The determinant $\begin{vmatrix} a &amp; b\\ c &amp; d \end{vmatrix}$ is define
\begin{equation*}
  \begin{vmatrix} a &amp; b\\ c &amp; d \end{vmatrix} = ad - bc
\end{equation*}
```

`matrix` with no bracket is an array without outer delimiters, so use `pmatrix` or `bmatrix` when you want brackets. `pmatrix` works both in text and in displays.

DOTS

COMMAND	LOOKS LIKE	USE
`\ldots`	$x_1, x_2, \ldots$	On the baseline, in a list
`\cdots`	$\cdots$	Centred, between operators
`\vdots`, `\ddots`	vertical and diagonal dots	In matrices
`\dotsc`, `\dotsb`, `\dotsm`, `\dotsi`	dots for commas, binary operators, products, integrals	Choose by what comes before and after

In a matrix, `\hdotsfor{4}` draws a row of dots spanning four columns. It takes an optional spacing argument, such as `\hdotsfor[2]{4}`.

DELIMITERS THAT GROW: `\LEFT` AND `\RIGHT`

```
\[
  \left( \frac{x}{y} \right)^{2} \qquad
  \left[ \sum_{k=1}^{n} k \right] \qquad
  \left\{ x \in \mathbb{R} : x &gt; 0 \right\}
\]
```

- Each `\left` needs a matching `\right`. Use `\right.` for an invisible closing delimiter.
- The delimiters need not match. A large brace on one side of a system of equations is a common use: `\left.` ... `\right\}`.
- Sometimes `\left` is too big. `\bigl`, `\bigr`, `\Bigl`, `\biggl`, `\Biggl` (and the `r` forms) give sizes you choose.

FRACTIONS AND BINOMIAL COEFFICIENTS

```
\frac{a+b}{c}          % fraction
\binom{n}{k}           % binomial coefficient
\tfrac{1}{2}           % smaller fraction, for use in a line
\dfrac{1}{2^{n-1}}     % larger fraction, even in text
\cfrac{1^2}{2+\cfrac{3^2}{2+\dotsb}}   % continued fraction
```

Inside text, `\frac` prints smaller than in a display. Use `\tfrac` or `\dfrac` to override that. `\dbinom` and `\tbinom` do the same for binomials.

`\genfrac{left}{right}{thickness}{style}{top}{bottom}` gives a custom fraction. Here, `\genfrac{\{}{\}}{0pt}{}{ij}{k}` prints the Christoffel symbol with braces, and `\genfrac{[}{]}{0pt}{}{ij}{k}` with square brackets. A zero thickness removes the line.

LIMITS ON LARGE OPERATORS

```
In text:      $\sum_{n=1}^{\infty} \frac{1}{n^2}$
In display:   \[ \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \]
Force text style in a line:   $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$
Limits at the side in display: $\sum\nolimits_{n=1}^{\infty}$
Limits above and below in text: $\sum\limits_{n=1}^{\infty}$
Integral with limits above and below: $\int\limits_0^{\infty}$
```

- In a display, the sum sign is larger and its limits sit above and below. In text, the limits sit at the side.
- `\limits` and `\nolimits` change the position. `\displaystyle` and `\textstyle` change the size.
- Integrals keep their limits at the side even in display, unless you add `\limits`.
- `\substack{i=1\\i\ne k}` puts two lines of subscript under one operator. `\sideset` puts symbols at all four corners.

ACCENTS, OVER AND UNDER

```
$\hat{a}$  $\bar{a}$  $\dot{a}$  $\ddot{a}$  $\tilde{a}$  $\vec{a}$
$\overset{\circ}{a}$      % a with a small circle above
$\underset{\circ}{a}$     % a with a small circle below
$\overbrace{a+b+c}^{\text{sum}}$   $\underbrace{x+y}_{\text{pair}}$
$\overrightarrow{AB}$     $\underline{ab}$
$\xrightarrow[\text{monic}]{f}$     % arrow with a label above and below
```

`\xrightarrow` and `\xleftarrow` (from amsmath) stretch to fit a label. The optional argument in square brackets is the label below.

8.5 New operators

Functions such as `\sin`, `\log` and `\lim` are already set upright. To make your own, declare it in the preamble:

```
\DeclareMathOperator{\cl}{cl}          % in the preamble
We denote the closure of $A$ in $Y$ by $\cl_Y(A)$.

\DeclareMathOperator*{\esup}{ess\,sup} % starred: limits above and below
$\|f\|_\infty=\esup_{x\in R}|f(x)|$
```

8.6 The many faces of mathematics: math fonts

COMMAND	EXAMPLE	USE
`\mathit`	x+y=z	Italic (default)
`\mathrm`	x+y=z (upright)	Upright text inside maths; the letter *d* in differentials
`\mathbf`	bold roman	Bold, without italics
`\mathsf`	sans serif	Matrices (ISO style, sans serif)
`\mathtt`	typewriter	Code-like symbols
`\mathcal`	calligraphic capitals	Like $\mathcal{L}$
`\boldsymbol`	bold italic	Vectors such as $\boldsymbol{a}+\boldsymbol{b}=\boldsymbol{c}$
`\mathbb`	blackboard bold	$\mathbb{R}$, $\mathbb{N}$ (needs amssymb)

`\mathbf` gives roman bold, not bold italic. For bold italic vectors use `\boldsymbol`. Note that `\boldsymbol{a+b=c}` makes the plus and equals bold too, because it affects every symbol in its braces.

> **ISO conventions (from the primer).** Simple variables are italic. Vectors are bold italic. The special numbers *e* and *i*, and the differential *d*, are upright. Define them once:

```
\newcommand{\me}{\mathrm{e}}
\newcommand{\mi}{\mathrm{i}}
\newcommand{\diff}{\mathrm{d}}   % then write $\diff x$
```

8.7 And that is not all

The primer points to the *amsldoc* document, which comes with TeX distributions, and to Knuth's *The TeXbook*, chapter 18, "Fine Points of Mathematics Typing", for more.

8.8 Symbols: the ones you need most

The primer has full symbol tables. Here are the most used ones.

GROUP	COMMANDS AND WHAT THEY PRINT
Greek lower case	`\alpha` α, `\beta` β, `\gamma` γ, `\delta` δ, `\epsilon` ε, `\varepsilon` ε, `\theta` θ, `\lambda` λ, `\mu` μ, `\pi` π, `\rho` ρ, `\sigma` σ, `\phi` φ, `\omega` ω
Greek capitals	`\Gamma` Γ, `\Delta` Δ, `\Theta` Θ, `\Lambda` Λ, `\Sigma` Σ, `\Pi` Π, `\Omega` Ω
Binary operators	`\pm` ±, `\times` ×, `\div` ÷, `\cdot` ·, `\circ` ∘, `\cap` ∩, `\cup` ∪, `\wedge` ∧, `\vee` ∨
Relations	`\le` ≤, `\ge` ≥, `\ne` ≠, `\approx` ≈, `\equiv` ≡, `\sim` ∼, `\subset` ⊂, `\subseteq` ⊆, `\in` ∈, `\propto` ∝
Arrows	`\to` or `\rightarrow` →, `\leftarrow` ←, `\Rightarrow` ⇒, `\Leftrightarrow` ⇔, `\mapsto` ↦, `\longrightarrow` ⟶
Miscellaneous	`\infty` ∞, `\partial` ∂, `\nabla` ∇, `\forall` ∀, `\exists` ∃, `\emptyset` ∅, `\neg` ¬, `\prime` ′, `\ldots` …
Variable-sized	`\sum` ∑, `\prod` ∏, `\int` ∫, `\oint`, `\bigcup`, `\bigcap`, `\bigoplus`
Log-like (upright)	`\sin`, `\cos`, `\tan`, `\log`, `\ln`, `\exp`, `\lim`, `\max`, `\min`, `\sup`, `\inf`, `\det`, `\gcd`, `\arg`
Delimiters	`(` `)` `[` `]` `\{` `\}` `\langle` `\rangle` `\lfloor` `\rfloor` `\lceil` `\rceil` `|` `\\|`
Accents	`\hat`, `\bar`, `\dot`, `\ddot`, `\tilde`, `\vec`, `\check`, `\breve`, `\grave`, `\acute`
Other constructions	`\widehat{abc}`, `\widetilde{abc}`, `\overline{abc}`, `\underline{abc}`, `\overbrace`, `\underbrace`, `\sqrt[n]{abc}`, `\frac{abc}{xyz}`

Packages for more symbols. Symbols marked "not predefined in LaTeX 2e" in the primer's tables need `amssymb` (or `latexsym`). That includes `\Box`, `\blacksquare`, `\nexists`, `\mathbb` and most of the AMS arrows and relations.

Remember (Part 8.4–8.8)

- `pmatrix` and `bmatrix` give brackets; `matrix` gives none.
- `\left` and `\right` must come in pairs. Use `.` to hide one side.
- `\frac` is small in text; use `\dfrac` to override it.
- `\DeclareMathOperator` goes in the preamble.
- For bold vectors use `\boldsymbol`, not `\mathbf`, when the symbols must be italic.